

Yifan Zhao

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EDUCATION

Colorado School of Public Health , M.S. Biostatistics & Informatics	Dec 2025
Worcester Polytechnic Institute , M.S. Bioinformatics & Computational Biology	2017
Worcester Polytechnic Institute , B.S. Bioinformatics & Computational Biology, with Distinction	2015

PEER REVIEWED PUBLICATIONS

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1. **Zhao Y**, Vestal BE, Moore CM. Evaluating reference-mixture matching in cell-type deconvolution with single-cell RNA-seq references. *Briefings in Bioinformatics*. 2026;27(2):bbag184. [DOI]
 2. King EM, **Zhao Y**, Moore CM, et al. Gpnmb and Spp1 mark a conserved macrophage injury response masking fibrosis-specific programming in the lung. *JCI Insight*. 2024;9(24):e182700. [DOI]
 3. Yin Y, Chen Y, Xu J, **Zhao Y**, et al. Molecular and spatial signatures of human and rat corpus cavernosum physiopathological processes at single-cell resolution. *Cell Reports*. 2024;43(9):114760. [DOI]
 4. Zhao LY, **Zhao YF**, Yao CC, et al. MHA, an interactive website for scRNA-seq data of male genitourinary development and disease. *Andrology*. 2023;11(6):1157–1162. [DOI]

RESEARCH EXPERIENCE

Graduate Research Assistant, CIQI Lab — Dr. Chuangqi Wang 2025–2026
University of Colorado Anschutz Medical Campus, Aurora, CO

- Analyzed dual scRNA-seq from infected human cell lines, evaluating bacterial-read reliability and methods for sparse host-pathogen signals.
- Improved guinea pig genome annotation using TOGA orthology inference and evaluated its effect on downstream single-cell analysis.

Graduate Research Assistant, Center for Innovative Design & Analysis — Dr. Mengli Xiao 2024–2025
Colorado School of Public Health, Aurora, CO

- Built a simulation framework for a causal-inference study and benchmarked meta-analysis methods.

Research Assistant, National Jewish Health — Drs. Tasha Fingerlin & Camille Moore Sep 2022–Jun 2024

- Analyzed bulk RNA-seq, methyl-seq, and WES data and communicated findings with experimental collaborators.

INDUSTRY EXPERIENCE

Bioinformatics Engineer, Shanghai Jingzhou Gene Technology Co., Ltd. Jan 2019–Apr 2022
Also known as Zhongke Purui, Shanghai, China

- Built scRNA-seq workflows for BD Rhapsody and 10× Genomics; trained colleagues in R, Linux, Git, and Anaconda.
- Led development of See-Cell, an R Shiny single-cell visualization tool, and contributed to the Male Health Atlas codebase.

Bioinformatics Engineer, Shanghai Liangtai Biotechnology Co., Ltd. Mar–Dec 2018

- Contributed to the development and testing of an exome RNA-seq analysis pipeline.

SELECTED QUANTITATIVE PROJECTS

- **Deep learning:** Used SHAP to interpret gene-level predictions from the DISSECT cell-type deconvolution model.
- **Single-cell simulation:** Assessed Wilcoxon rank-sum performance under unimodal and multimodal expression scenarios.
- **Genomics:** Compared RNA-seq differential-expression workflows under sample-size imbalance.
- **Genetic association:** Compared clumping-and-thresholding PRS with LASSO, ridge, and elastic-net models for idiopathic pulmonary fibrosis.

TEACHING AND ACADEMIC SERVICE

Teaching Assistant, Advanced Biostatistical Methods I (BIOS 6618) Fall 2024
Grader, Bioinformatics (BSBT 6805) Spring 2025
Statistical Genomics Working Group, organizer, president, and treasurer 2023–2026

TECHNICAL SKILLS

Genomics: single-cell and bulk RNA-seq, differential expression, cell-type deconvolution, methyl-seq, WES, comparative genomics
Statistics: longitudinal and survival analysis, generalized linear and mixed models, causal inference, meta-analysis, simulation
Computing: R, R Shiny, Python, Linux/Bash, Git, Anaconda, AI-assisted software development